Divisor Analysis

Time limit: 1.00 s
Memory limit: 512 MB

CSES 2182 | Mathematics | https://cses.fi/problemset/task/2182

Given an integer, your task is to find the number, sum and product of its divisors. As an example, let us consider the number 12:

- the number of divisors is 6 (they are 1, 2, 3, 4, 6, 12)
- the sum of divisors is 1+2+3+4+6+12=28
- the product of divisors is $1 \cdot 2 \cdot 3 \cdot 4 \cdot 6 \cdot 12 = 1728$

Since the input number may be large, it is given as a prime factorization.

Input

The first line has an integer n: the number of parts in the prime factorization.

After this, there are n lines that describe the factorization. Each line has two numbers x and k where x is a prime and k is its power.

Output

Print three integers modulo 10^9+7: the number, sum and product of the divisors.

Constraints

- $1 \le n \le 10^5$
- $2 \le x \le 10^6$
- each x is a distinct prime
- $1 \le k \le 10^9$

Example

Input
2 2 2 3 1

Output
6 28 1728